

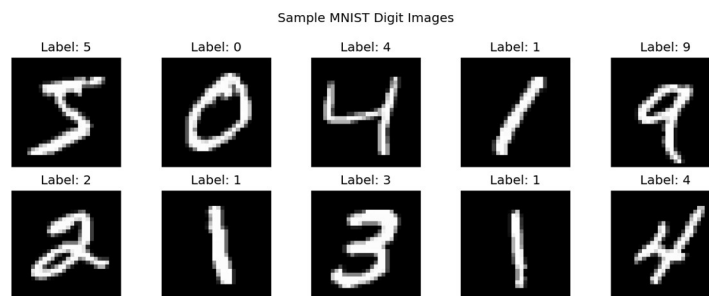
NAVIOTECH SOLUTION PVT LTD

Cybersecurity & AI Internship Program

PROJECT REPORT

Handwritten Digit Recognition Using Neural Networks

(MNIST Dataset)



Submitted by

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Submitted to
Naviotech Solution Pvt Ltd

Certificate

This is to certify that K. Ganga Siva Kumar Reddy, a Cybersecurity Internship student from Atria University, has successfully completed the project titled "Handwritten Digit Recognition Using Neural Networks (MNIST Dataset)" as part of the internship program at Naviotech Solution Pvt Ltd.

The project was carried out under the guidance and supervision of the assigned project mentor at Naviotech Solution Pvt Ltd, and reflects the student's understanding of Artificial Intelligence, Machine Learning, and Neural Network concepts applied to a real-world image classification problem.

This report is submitted in partial fulfilment of the requirements of the internship program.

Date: 23/07/2026

Place: Bengaluru

K. Ganga Siva Kumar Reddy

Signature of Student

Ameen MD

Signature of Mentor / Authority

Acknowledgement

I would like to express my sincere gratitude to Naviotech Solution Pvt Ltd for providing me the opportunity to undertake this internship and work on this Artificial Intelligence project. This experience has significantly enhanced my practical understanding of Neural Networks and their application in image classification.

I am thankful to my project mentor and the technical team at Naviotech Solution Pvt Ltd for their continuous guidance, valuable suggestions, and support throughout the duration of this project.

I also extend my gratitude to the faculty of Atria University for their academic support and encouragement, which enabled me to apply my classroom learning to a real industry-oriented project.

Finally, I would like to thank my classmates and peers who participated in internship project activities and provided helpful discussions during the course of this work.

Abstract

Handwritten digit recognition is a foundational problem in the field of Artificial Intelligence and Computer Vision, with widespread applications in banking, postal services, and automated data entry systems. This project, "Handwritten Digit Recognition Using Neural Networks," presents the design, development, and evaluation of a Convolutional Neural Network (CNN) model trained on the MNIST dataset to classify handwritten digits ranging from 0 to 9.

The MNIST dataset, comprising 70,000 grayscale images of size 28x28 pixels (60,000 for training and 10,000 for testing), was preprocessed through normalization and reshaping before being fed into a CNN architecture consisting of convolutional, pooling, dense, and dropout layers. The model was trained using the Adam optimizer and categorical cross-entropy loss function over multiple epochs.

Upon evaluation, the trained model achieved a test accuracy of 99.24%, demonstrating strong generalization capability on unseen handwritten digit images. Performance was further validated using a confusion matrix, precision, recall, and F1-score metrics across all ten digit classes. The project provides hands-on exposure to the complete machine learning pipeline: data collection, preprocessing, model design, training, evaluation, and prediction.

This work reinforces the practical relevance of deep learning techniques in real-world OCR (Optical Character Recognition) systems and lays the foundation for more advanced computer vision applications.

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1. Introduction

1.1 Background

Artificial Intelligence and Machine Learning have transformed the way computers interpret visual information. Among the many applications of AI, image classification remains one of the most researched and widely deployed capabilities. Handwritten digit recognition, in particular, has served as a benchmark problem for evaluating machine learning and deep learning models since the introduction of the MNIST dataset by Yann LeCun and his collaborators.

The ability of a machine to correctly interpret handwritten numerals has direct real-world utility — from reading handwritten bank cheques to sorting postal mail by ZIP code. This project explores how Neural Networks, specifically Convolutional Neural Networks (CNNs), can be trained to solve this classification task with high accuracy.

1.2 Problem Statement

Manually verifying or transcribing handwritten digits is time-consuming and error-prone when performed at scale. There is a need for an automated, accurate, and efficient system capable of recognizing handwritten digits from scanned or photographed images. This project addresses that need by developing a Neural Network-based digit classification system.

1.3 Objectives

- Understand the fundamentals of Neural Networks and Convolutional Neural Networks.
- Learn and apply image classification techniques to a real dataset.
- Train a Neural Network model using the MNIST dataset.
- Evaluate model accuracy and performance using standard classification metrics.
- Deploy a simple digit recognition and prediction system.

1.4 Scope of the Project

The scope of this project is limited to classifying isolated, pre-segmented handwritten digit images (0-9) from the standard MNIST dataset. The project covers the complete pipeline from data acquisition to model deployment for prediction, but does not extend to full-page document OCR or multi-digit sequence recognition.

2. Literature Review

Handwritten digit recognition has been extensively studied since the 1990s. LeCun et al. introduced the LeNet-5 architecture, one of the earliest successful applications of Convolutional Neural Networks to digit recognition, achieving high accuracy on the MNIST dataset and laying the groundwork for modern deep learning-based computer vision.

Subsequent research explored various machine learning approaches to this problem, including Support Vector Machines (SVMs), k-Nearest Neighbors (k-NN), and traditional fully-connected Neural Networks. While these methods achieved reasonable accuracy, they were generally outperformed by CNN-based architectures, which are specifically designed to capture spatial hierarchies in image data through convolutional and pooling operations.

With the advent of deep learning frameworks such as TensorFlow and Keras, building and training CNNs has become significantly more accessible. Modern CNN architectures routinely achieve accuracy exceeding 99% on the MNIST benchmark, making it a standard dataset for validating new architectures, optimizers, and regularization techniques such as Dropout and Batch Normalization.

This project builds upon these established techniques, implementing a compact CNN architecture suitable for educational and practical demonstration purposes while achieving performance comparable to published benchmarks.

3. Methodology

3.1 Dataset Description

The MNIST (Modified National Institute of Standards and Technology) dataset was used for this project. It consists of 70,000 grayscale images of handwritten digits (0-9), each of size 28x28 pixels.

Attribute	Value
Total Images	70,000
Training Samples	60,000
Testing Samples	10,000
Image Size	28 x 28 pixels (grayscale)
Number of Classes	10 (digits 0-9)

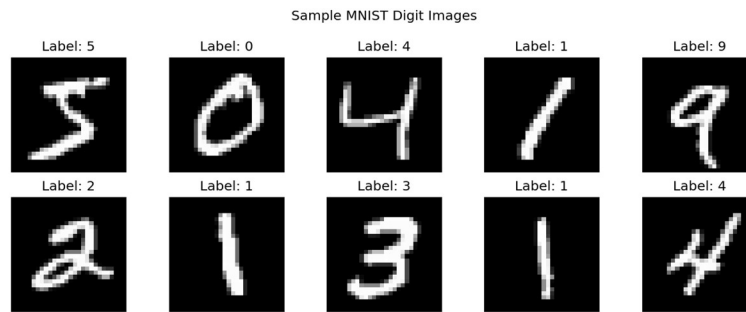


Figure 3.1: Sample images from the MNIST dataset

3.2 Technologies Used

- Programming Language: Python
- Libraries: NumPy, Pandas, Matplotlib, Seaborn, TensorFlow, Keras, Scikit-learn
- Development Environment: Jupyter Notebook / VS Code

3.3 Data Preprocessing

The raw pixel values (ranging from 0-255) were normalized to a range of 0-1 by dividing by 255, which helps the neural network converge faster during training. The images were reshaped to include a single channel dimension (28x28x1) to be compatible with the Conv2D input layer. Labels were one-hot encoded into 10-dimensional vectors representing each digit class.

4. System Design

4.1 Model Architecture

A Convolutional Neural Network (CNN) was designed with the following architecture:

Layer	Output Shape	Parameters
Conv2D (32 filters, 3x3, ReLU)	(26, 26, 32)	320
MaxPooling2D (2x2)	(13, 13, 32)	0
Conv2D (64 filters, 3x3, ReLU)	(11, 11, 64)	18,496
MaxPooling2D (2x2)	(5, 5, 64)	0
Flatten	(1600)	0
Dense (128, ReLU)	(128)	204,928
Dropout (0.3)	(128)	0
Dense (10, Softmax)	(10)	1,290

Total Trainable Parameters: 225,034

4.2 System Workflow

- Phase 1 - Data Collection: Load the MNIST dataset and explore sample images.
- Phase 2 - Data Preprocessing: Normalize pixel values and reshape image dimensions.
- Phase 3 - Model Development: Design the CNN architecture with convolution, pooling, and dense layers.
- Phase 4 - Model Training: Train the model using the training set with validation split.
- Phase 5 - Testing and Evaluation: Evaluate the trained model on the unseen test set.
- Phase 6 - Prediction: Use the trained model to predict digits on new/unseen images.

4.3 Compilation Settings

- Optimizer: Adam
- Loss Function: Categorical Cross-Entropy
- Evaluation Metric: Accuracy
- Epochs: 5
- Batch Size: 128

5. Implementation

The model was implemented in Python using TensorFlow and Keras. The complete implementation follows the six-phase workflow described in the System Design section. Key implementation steps included building the Sequential CNN model, compiling it with the Adam optimizer, and training it for 5 epochs with a 10% validation split to monitor generalization performance during training.

5.1 Training Progress

The model was trained over 5 epochs, with training and validation accuracy/loss monitored at each epoch:

Epoch	Train Accuracy	Train Loss	Val Accuracy	Val Loss
1	91.95%	0.2672	98.18%	0.0616
2	97.55%	0.0794	98.70%	0.0404
3	98.23%	0.0565	98.77%	0.0405
4	98.59%	0.0438	98.70%	0.0392
5	98.91%	0.0361	99.12%	0.0299

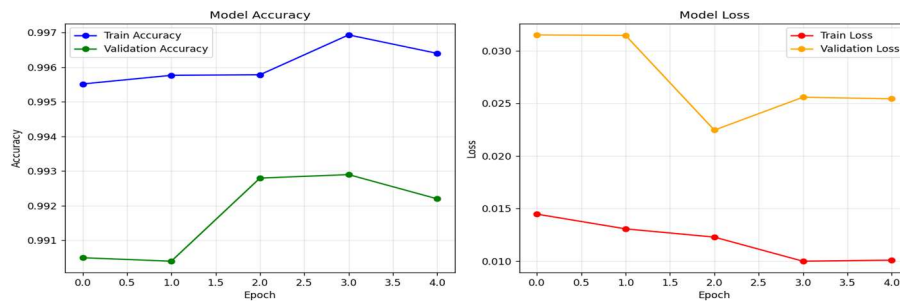


Figure 5.1: Model training accuracy and loss curves over 5 epochs

6. Results and Discussion

6.1 Overall Performance

The final trained model achieved the following performance on the held-out test set of 10,000 images:

Metric	Value
Test Accuracy	99.24%
Test Loss	0.0225

6.2 Classification Report

The precision, recall, and F1-score for each digit class (0-9) were computed as follows:

Digit	Precision	Recall	F1-Score
0	0.99	1.00	0.99
1	0.99	1.00	1.00
2	0.99	0.99	0.99
3	0.99	0.99	0.99
4	0.99	0.99	0.99
5	0.98	0.99	0.99
6	1.00	0.98	0.99
7	0.99	0.99	0.99
8	0.99	0.99	0.99
9	0.98	0.99	0.99

6.3 Confusion Matrix

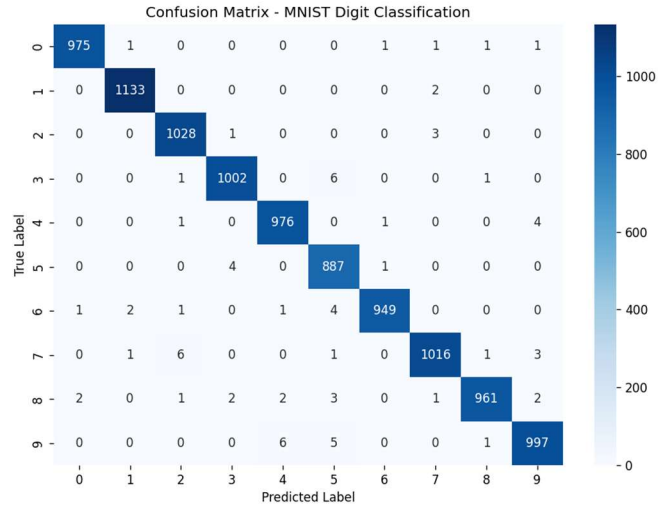


Figure 6.1: Confusion matrix showing predicted vs actual digit classes

6.4 Sample Predictions

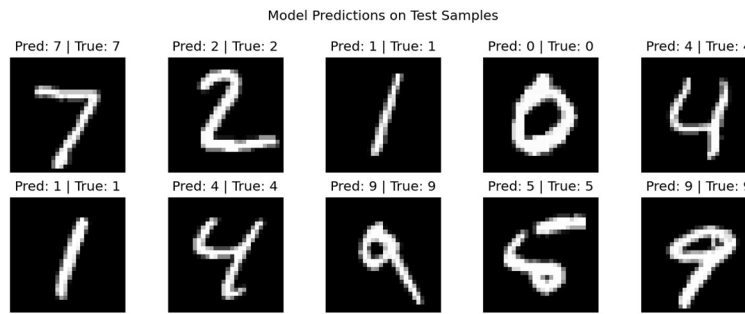


Figure 6.2: Model predictions compared against true labels on test samples

6.5 Discussion

The model achieved a test accuracy of 99.24%, exceeding the expected outcome benchmark of 95%. The confusion matrix shows that most misclassifications occurred between visually similar digits, such as 4 and 9, or 3 and 5, which is consistent with known challenges in handwritten digit recognition. The high precision and recall values across all classes indicate that the model generalizes well and does not exhibit significant bias toward any particular digit.

7. Conclusion

This project successfully demonstrates the application of Convolutional Neural Networks for handwritten digit recognition using the MNIST dataset. By following a structured machine learning pipeline — data collection, preprocessing, model development, training, evaluation, and prediction — a highly accurate digit classification model (99.24% test accuracy) was developed.

The project provided practical, hands-on experience with key AI/ML concepts including convolutional layers, pooling, dropout regularization, and performance evaluation using confusion matrices and classification metrics. It reinforces the relevance of deep learning in solving real-world image classification problems such as OCR, automated form processing, and postal code recognition.

8. Future Scope

- Extend the model to recognize multi-digit sequences and full alphanumeric characters.
- Deploy the model as a web or mobile application for real-time digit recognition.
- Experiment with deeper architectures (ResNet, EfficientNet) to further improve accuracy.
- Apply data augmentation techniques to improve robustness against varied handwriting styles.
- Integrate the model into real-world OCR pipelines such as cheque processing and postal automation systems.

9. References

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